Lesson 1.3

Midpoint & Distance Formula

Two formulas — and how to never mix them up

The trick

Midpoint = Mean

The trick

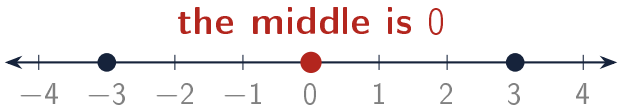
Midpoint = Mean

Distance = Difference

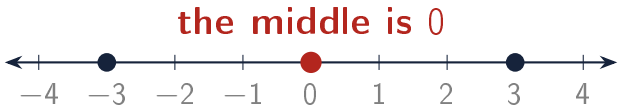
Midpoint: simple case first



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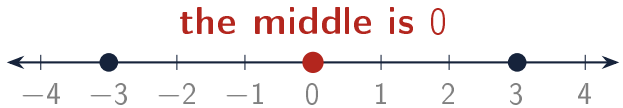


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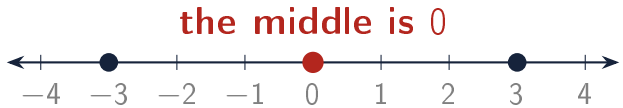
Add and halve: $\dfrac{-3+3}{2} = 0$ ✓

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Add and halve: $\dfrac{-3+3}{2} = 0$ ✓ Subtract: $\dfrac{3-(-3)}{2} = 3$ ×

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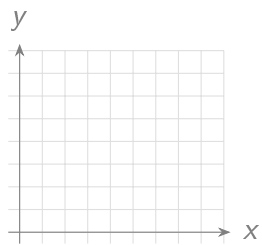


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So the midpoint **adds** — it's the **average** (mean).

Midpoint formula

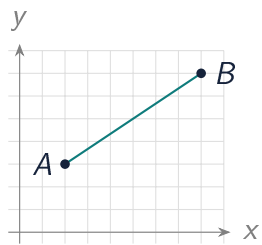
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Example: $A(2, 3)$, $B(8, 7)$

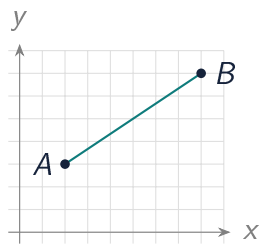


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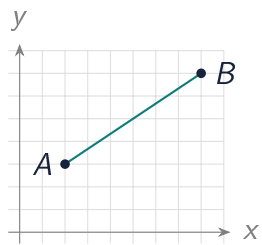
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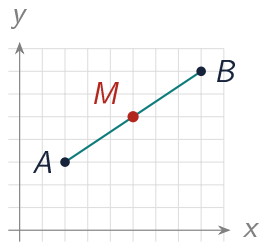
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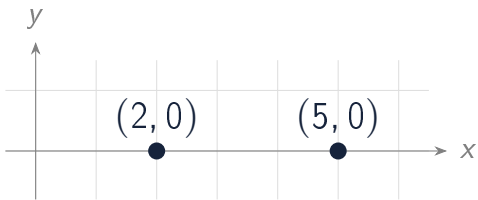
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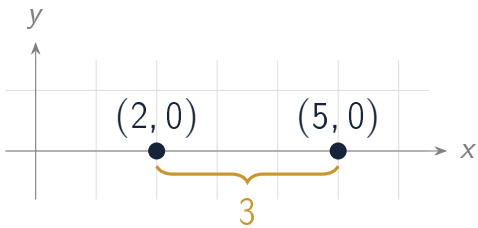
$M = (5, 5)$



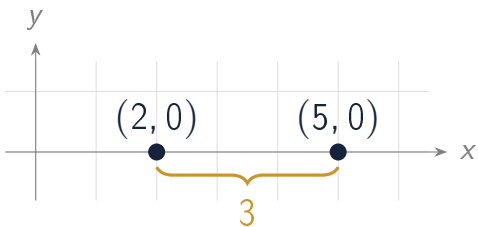
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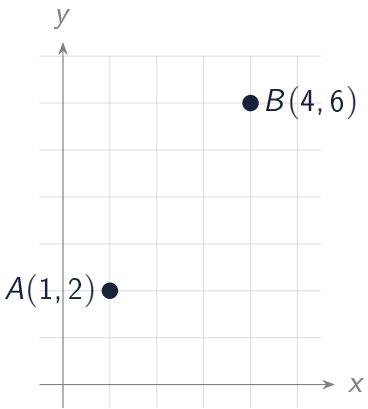


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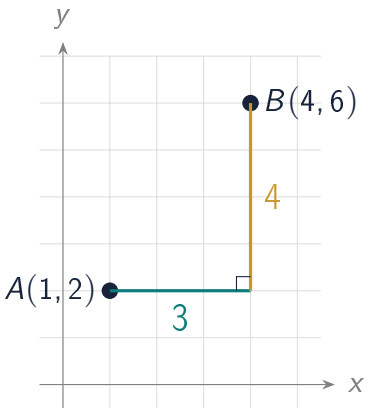
Count: 3. That's $5 - 2$ — a **difference**.

Distance = Pythagoras on a grid



Find AB.

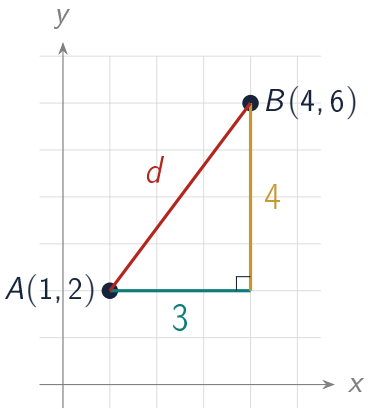
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Find AB.

Across 3, up 4.

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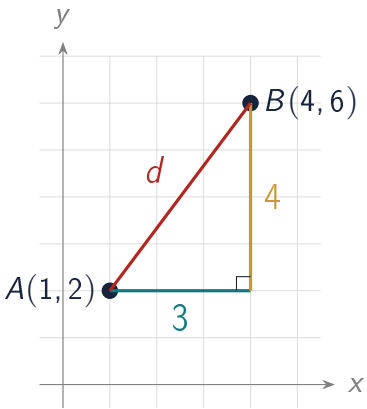


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Across 3, up 4.

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$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Two common mistakes

Watch out! Square **each** difference before adding:
$\sqrt{3^2 + 4^2} = 5$, not $3 + 4 = 7$.

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Watch out! Testing add vs. subtract? Don't use 0 and 2 — both give 1. Use 2 and 4: the middle is $3 = \frac{2+4}{2}$.

Your turn! Pause the video and try it.

$P(-1, 4)$ and $Q(5, -4)$. Find the midpoint of $\overline{PQ}$ and the length PQ.

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$$PQ = \sqrt{(5-(-1))^2 + (-4-4)^2} = \sqrt{6^2+8^2} = \sqrt{100} = 10$$

Remember

- **Midpoint = Mean**: add the coordinates, divide by 2.
- **Distance = Difference**: subtract, square, add, square root.
- The distance formula is just the Pythagorean Theorem.

Now practice at your level:

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